

Research Article

Multi-dimensional search for drug–target interaction prediction by preserving the consistency of attention distribution

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ARTICLE INFO

Keywords:

Drug–target interaction
Attention mechanism
Drug decomposition
Multi-dimensional search
Consistency

ABSTRACT

Predicting drug–target interaction (DTI) is a crucial step in the process of drug repurposing and new drug development. Although the attention mechanism has been widely used to capture the interactions between drugs and targets, it mainly uses the Simplified Molecular Input Line Entry System (SMILES) and two-dimensional (2D) molecular graph features of drugs. In this paper, we propose a neural network model called MdDTI for DTI prediction. The model searches for binding sites that may interact with the target from the multiple dimensions of drug structure, namely the 2D substructures and the three-dimensional (3D) spatial structure. For the 2D substructures, we have developed a novel substructure decomposition strategy based on drug molecular graphs and compared its performance with the SMILES-based decomposition method. For the 3D spatial structure of drugs, we constructed spatial feature representation matrices for drugs based on the Cartesian coordinates of heavy atoms (without hydrogen atoms) in each drug. Finally, to ensure the search results of the model are consistent across multiple dimensions, we construct a consistency loss function. We evaluate MdDTI on four drug–target interaction datasets and three independent compound–protein affinity test sets. The results indicate that our model surpasses a series of state-of-the-art models. Case studies demonstrate that our model is capable of capturing the potential binding regions between drugs and targets, and it shows efficacy in drug repurposing. Our code is available at <https://github.com/lhhu1999/MdDTI>.

1. Introduction

In the biopharmaceutical industry, drug discovery is extremely time-consuming and costly, while the success rate is only 2.01% (Yeu et al., 2015). The development of a new drug takes about 10–15 years (Wong et al., 2014) and costs more than \$2.6 billion (Mullard, 2014). In contrast, drug repurposing is favored by researchers because of its low risk, shorter development time and less investment. Studying the drug–target interaction (DTI) is a crucial step in drug repurposing. In addition, quickly judging the binding site between drugs and targets can also speed up the development of new drugs. Traditionally, the binding affinities between drugs and targets can be measured through high-throughput screening experiments, but it is time-consuming (Cohen, 2002).

Recently, many models based on machine learning have been proposed to predict DTI (Chen et al., 2016) or drug–target affinity (DTA). Tsubaki et al. (2019) designed an end-to-end model based on the attention mechanism. The model extracts the features of proteins and drugs through convolutional neural networks (CNNs) (Krizhevsky

et al., 2017) and graph neural networks (GNNs) (Scarselli et al., 2008) respectively, which can achieve better performance on the unbalanced datasets. Wang et al. (2021b) introduced amino acid residues that can directly bind to ligands into protein–ligand affinity prediction. Then, the dilated convolution and traditional convolution are used to capture both long-range and short-range interactions. Nguyen et al. (2021) proposed a DTA prediction model based on a combination of GNNs and CNN, which uses graph convolutional network (GCN) (Kipf and Welling, 2016), graph attention network (GAT) (Veličković et al., 2017), graph isomorphism network (GIN) (Xu et al., 2018) and GAT-GCN to extract the drug features, respectively. MRBDA (Zhang et al., 2022) uses an encoder level skip connection to construct a molecular representation block, which can effectively capture the interaction sites between drugs and proteins. Moltrans (Huang et al., 2021) divides the drug and target into segments of different lengths by frequent consecutive sub-sequence (FCS) mining algorithm, and then capture the chemical semantics of sub-structures through an augmented transformer module. The model can narrow the search range of binding

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<https://doi.org/10.1016/j.compbiolchem.2023.107968>

Received 6 August 2023; Received in revised form 27 September 2023; Accepted 5 October 2023

Available online 7 October 2023

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sites between drugs and proteins in practical applications. However, some amino acid sequences that have a low frequency but a greater effect on protein may be forcibly disassembled. Sun et al. (2021) proposed an autoencoder-based model, which uses similarity to build a new loss function to ensure the consistency of spatial distribution of drug features. Ru et al. (2022) treats the DTA prediction as a search ranking task. This method can use the characteristics of learning sorting to set different queries, so that the model can be applied to many different scenarios. There are also some prediction models based on heterogeneous networks (Chen et al., 2012; Yan et al., 2019; An and Yu, 2021; Zhou et al., 2021; Su et al., 2022; Li et al., 2022c). These heterogeneous network models also incorporated the drug–drug associations, drug–side-effects associations, drug–disease associations and protein–disease associations.

However, the aforementioned methods mainly use the Simplified Molecular Input Line Entry System (SMILES) (Wang et al., 2021b; Zhou et al., 2021; Zhao et al., 2022; Sivangi et al., 2023) or two-dimensional (2D) molecular graph (Tsubaki et al., 2019; Nguyen et al., 2021; Li et al., 2022a; Wang et al., 2021a) of drugs to extract features. In addition, heterogeneous networks mainly predict the relationship between drugs and targets by learning the potential associations between drugs and other types of data. Therefore, methods based on heterogeneous networks usually do not analyze the structure of molecules. In fact, drugs have different forms in space due to the effect of interatomic force and chemical bonds. The spatial form of the drug is essential to the formation of binding pocket. Therefore, it is crucial to effectively utilize the spatial coordinates of atoms for predicting DTI.

In this paper, we propose a drug–target interaction prediction model, named MdDTI. Our model innovatively predicts the binding sites between drugs and targets from the multiple dimensions of drug structure, namely the 2D substructures and the three-dimensional (3D) spatial structure. Since both prediction methods aim to identify the binding sites between the same drug and target, there should be some degree of similarity between the results obtained from these two methods. Therefore, we constructed a consistency loss function that ensures the similarity between the two results by adjusting the attention distribution of the drug on target residues. We used seven datasets to evaluate the performances of our models. The results indicate that our models perform better than the baseline models on multiple metrics. Our innovations are summarized as below:

1. We design a decomposition strategy based on the 2D structure of drugs to generate the Improved Chemistry-inspired Molecular Fragments (ICMF). Compared to the SMILES-based method, the Explainable Substructure Partition Fingerprint (ESPF) (Huang et al., 2019), our method does not generate redundant descriptor fragments.
2. We propose a method which effectively utilizes the spatial Cartesian coordinates of heavy atoms to build the 3D spatial feature expression matrix for drugs. For hydrogen atoms in molecules, we found that using GNNs to facilitate information exchange between all atoms in molecules first, and then removing the features of the hydrogen atoms can enhance the robustness of the model.
3. We construct a consistency loss function to ensure the consistency of the model's attention distribution towards the target across both the dimensions of 2D substructure and 3D spatial structure.

2. Datasets

In this work, we collected a total of seven datasets for testing. Among them, there were four datasets for drug–target interaction (DTI) and three datasets for compound–protein binding affinity (CPA). The DTI datasets are used to predict whether there are interactions between drugs and targets. The CPA datasets are used to predict the interaction strength between compounds and proteins.

Table 1

Summary of the four DTI datasets used in this study.

Datasets	Drug	Target	Interaction	Positive	Negative
Human	2726	2001	6728	3364	3364
C.elegans	1767	1876	7786	3893	3893
Davis	68	379	25772	7320	18452
KIBA	2068	225	116350	22154	94196

Table 2

Summary of the three CPA datasets used in this study.

Datasets	Compound	Protein	Train	Test
Kd	5895	812	8778	3811
EC50	31970	513	26413	11483
Ki	93437	1619	101134	43391

The detailed information of the four DTI datasets Human (Liu et al., 2015), C.elegans (Liu et al., 2015), Davis (Zhao et al., 2022; Davis et al., 2011) and KIBA (Zhao et al., 2022; Tang et al., 2014) is summarized in Table 1. In the Human and C.elegans datasets, the number of positive samples (with a label of 1) and negative samples (with a label of 0) is the same. The Davis and KIBA datasets were originally constructed by Davis et al. (2011) and Tang et al. (2014), respectively. Zhao et al. (2022) removed the duplicated drug–target pairs from these two datasets.

The detailed information of the three CPA datasets Kd, EC50 and Ki (Karimi et al., 2019) is summarized in Table 2. The affinity refers to the binding capacity between compounds and proteins, which is usually measured by wet experiments. The evaluation index usually includes dissociation constant (K_d), half maximal effective concentration (EC_{50}), inhibition constant (K_i), half maximal inhibition concentration (IC_{50}), etc.

Since drugs belong to the category of compounds and targets are amino acid sequences, DTI and CPA can be uniformly described as (D , T , I_s), where D represents drug or compound, and T represents target or protein. For DTI, the I_s can only take a value of either 0 or 1, while for CPA, the I_s can have any value greater than or equal to 0.

3. Methods

The detailed structure of MdDTI is shown in Fig. 1. Our model mainly consists of four parts: (1) 2D substructures extraction of drugs. We compared the performance of our proposed method, ICMF, which is based on 2D structure, with the traditional method ESPF, which relies on 1D SMILES representation. (2) 3D features extraction of drugs. We introduce the spatial Cartesian coordinates of heavy atoms and propose a method to generate the spatial expression matrix of drugs. (3) Target search network. The target search network uses the attention mechanism to search the binding sites from the two dimensions of drug structure. Then, we built a consistency loss function based on attention scores. (4) Interaction prediction network. This part is a multi-layer perceptual (MLP), which is used to output the final prediction results.

Note that the input SMILES is unified using RDKit (Landrum, 2013) (by converting SMILES to a mol object using RDKit, and then converting the mol object back to unified SMILES). Because the SMILES representation of the same drug may vary, standardizing it facilitates easier processing. Furthermore, we have summarized the symbols that facilitate understanding in Table 3.

3.1. 2D substructures extraction of drugs

Inspired by Zhang et al. (2021), we designed the ICMF. Unlike the ESPF algorithm, our method is based on the 2D molecular structure and does not generate descriptor fragments. Descriptor fragments refer to the meaningless segments generated after breaking down the SMILES

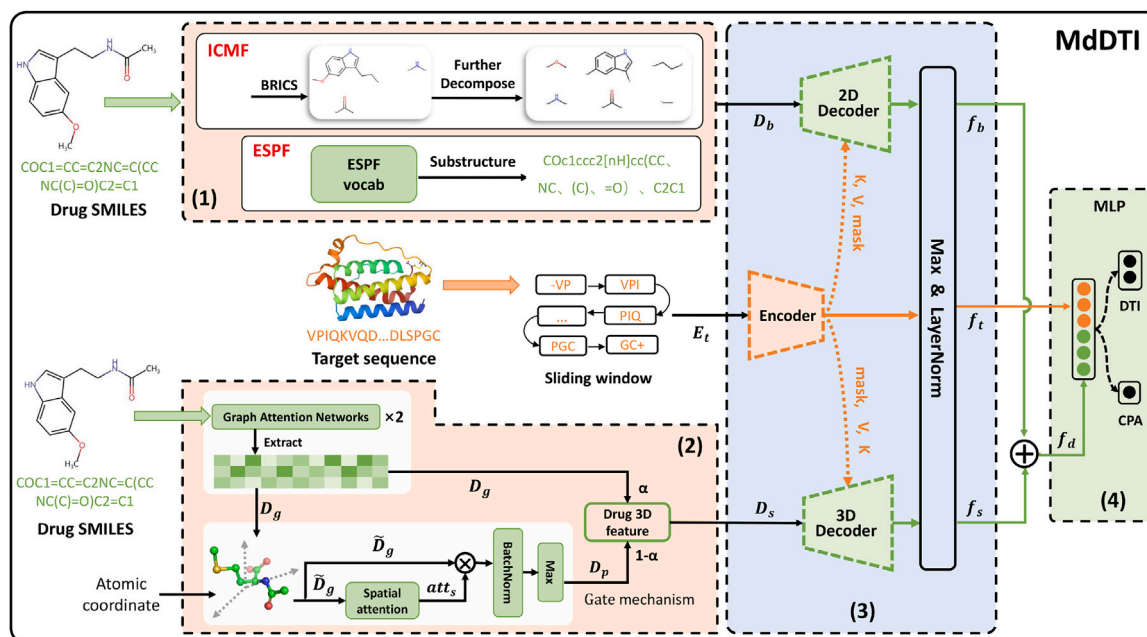


Fig. 1. The overall architecture of MdDTI. The inputs of the MdDTI are the SMILES string of drugs and the amino acid sequence of targets. The model has four main parts: (1) The 2D substructures extraction of drugs. ICMF and ESPF are two methods to decompose drugs. (2) The 3D features extraction of drugs. The dimensional level gate mechanism is used to fuse the local features and spatial structure features of the drug. (3) Target search network. (4) Interaction prediction network, which is a three-layer perceptron. The '-' and '+' in the sliding window are used to represent the start and end of the amino acid sequence.

Table 3
Symbols used in the article and their meanings.

Symbol	Meaning
l'	The length of the drugs before compression.
l	The length of the drugs after compression.
p	The number of 2D substructures in the drugs.
q	The length of the targets.
d	The dimension of the embedding.
k	The number of attention heads.
n	The size of the batch.

string of a drug, such as '(' and ')'. These redundant descriptor fragments do not contain any atom or structural information relevant to the drug. In contrast, ESPF directly divides the SMILES string into different fragments, which can easily generate some redundant descriptor fragments.

The idea of the ESPF algorithm is similar to that of Byte Pair Encoding (BPE) algorithm (Gage, 1994). Huang et al. (2019) collected approximately 2 million SMILES and extracted substructures that appeared with a frequency higher than 100. Finally, they obtained approximately 23,000 substructures. During usage, the SMILES to be decomposed is compared with the obtained substructure dataset, and the highest frequency substructure is sequentially extracted from the SMILES. This method can successfully identify some substructures that have an impact on the chemical properties of drugs, such as carboxyl group and benzene ring.

Fig. 2 is an example flowchart illustrating the decomposition of a drug (ChEMBL1986265) using the ICMF algorithm. The ICMF method decomposes the 2D molecular graph of drugs through five steps:

1. We use the Breaking of Retrosynthetically Interesting Chemical Substructures (BRICS) (Degen et al., 2008) algorithm to decompose the drug. BRICS to decompose drugs from the perspective of chemical reaction, which retains valuable substructure as much as possible. However, the number of substructures decomposed by BRICS is too small, which is disadvantageous to searching binding sites between drugs and targets. Therefore, it is necessary to further decompose the substructures.

2. Due to the small number of substructures decomposed by BRICS, which is disadvantageous to searching binding sites between drugs and targets. Therefore, it is necessary to further decompose the substructures. We established two chemical bond breaking rules. The first rule involves breaking a bond between an atom in a ring and an atom not in a ring. The second rule involves breaking all single bonds if an atom is surrounded by more than two bonds.
3. Search for functional groups, such as carboxyl groups, that may be incorrectly decomposed in step 2.
4. Remove some fragments that were incorrectly decomposed in step 2, and then merge the extracted functional groups and the cleaned molecular fragments into the final set of substructures.
5. Sort molecular fragments based on the index of atoms in the fragments. The first substructure consists of 3 heavy atoms (C, C, O), with indices 0, 1, and 2 in the drug. Therefore, this fragment is represented by the minimum index 0. Similarly, the second substructure is represented by the minimum index 3. Finally, all substructures are arranged in ascending order based on their indices. From the sorting results, our sorting strategy for molecular fragments is essentially a depth-first search algorithm.

3.2. 3D features extraction of drugs

To increase the encodable range, each atom in the drug is represented using the local structure type (including the key type, atom type and graph structure) within the range of its two chemical bonds. In the experiment, the bond type is identified directly through RDKit, such as single bonds, double bonds and triple bonds. Besides, we distinguish aromatic hydrocarbon atoms separately. For example, aromatic carbon atoms and non-aromatic carbon atoms are categorized as two different types of atoms.

3.2.1. Aggregating local features by GAT

GAT can control the influence of each atom by assigning different weights to its adjacent atoms, and then update them through continuous aggregation. In the experiment, we used GAT aggregation to update twice. If we use $h_i \in \mathbb{R}^d$ to represent the embedding vector of the

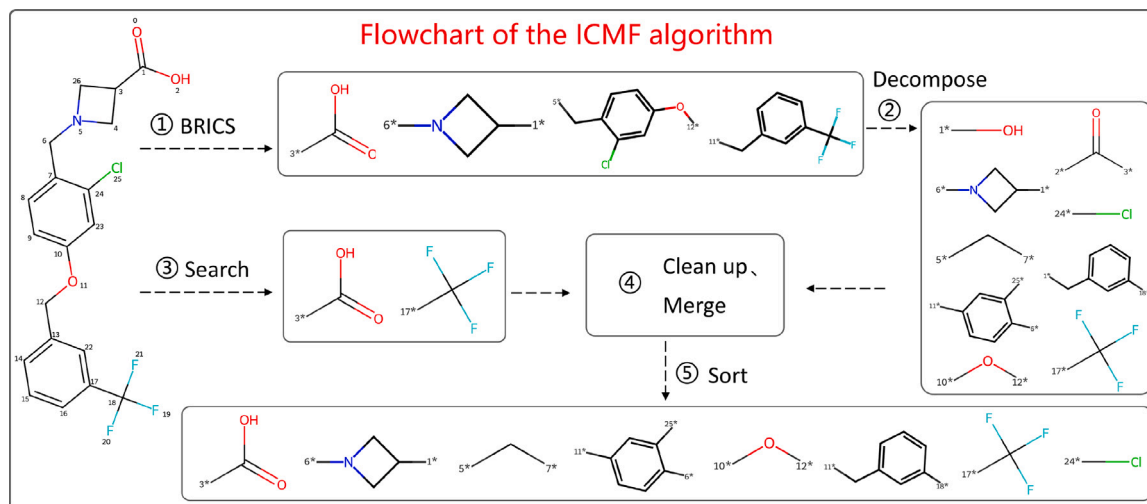


Fig. 2. Flowchart of the ICMF algorithm for decomposing the drug (CHEMBL1986265). ① The BRICS algorithm is used for the initial decomposition of the drug. ② The obtained drug substructures are further decomposed using custom rules. ③ Identify functional groups that might be incorrectly decomposed. ④ Remove the mistakenly decomposed molecular fragments. Then, merge the extracted functional groups and the cleaned molecular fragments as the final set of substructures. ⑤ Sort molecular fragments based on the index of atoms in the fragments.

i th atom and $A \in \mathbb{R}^{l' \times l'}$ represents the adjacency matrix, where l' represents the number of atoms in the drug. The formula for calculating the attention score α_{ij} between the i th atom and the j th atom is as follows:

$$\alpha_{ij} = R(\sigma(a^T [Wh_i \parallel Wh_j]), A_{ij}) \quad (1)$$

$$R(x, \beta) = \begin{cases} x, & \text{if } \beta = 1 \\ \epsilon, & \text{if } \beta = 0 \end{cases} \quad (2)$$

where $\parallel$ is the concatenation operation, W is a training parameter matrix, $a \in \mathbb{R}^{2d \times 1}$ is a shared attention mechanism and σ is the LeakyReLU activation function. R replaces the attention between non-adjacent atoms with a very small value ϵ , such as $-9e15$ (the attention will change to near-zero after applying *softmax*). Then, the atom features d_i are updated based on the attention to neighboring atoms:

$$d_i = \text{softmax}(\alpha_i) \cdot Wh \quad (3)$$

3.2.2. Extracting features of heavy atoms in drug

Previous researchers have primarily used two methods to handle hydrogen atoms in drugs. The first method involves discarding hydrogen atoms during preprocessing, while the second method involves using all atoms throughout the entire process. However, directly removing hydrogen atoms will inevitably have an impact on the extraction of molecular features. Additionally, due to the significant proportion of hydrogen atoms in molecules, using hydrogen atoms throughout the process would lead to a more redundant feature map, which is detrimental to model training.

We opt for a compromise solution, where in the step 1, we utilize all atoms to aggregate neighbor information, and then compress the feature map by removing the feature vectors of hydrogen atoms. If we use $D_i = \{d_1, d_2, \dots, d_{l'}\} \in \mathbb{R}^{l' \times d}$ to represent the feature matrix of the i th drug in a batch. $\Phi_i = \{\varphi_1, \varphi_2, \dots, \varphi_{l'}\} \in \mathbb{R}^{l'}$ represents the indicator bits for heavy atoms of i th drug and $\Theta_i = \{\theta_1, \theta_2, \dots, \theta_{l'}\} \in \mathbb{R}^{l'}$ represents the indicator bits for the padding nodes of i th drug. When the i th atom is a heavy atom, $\varphi_i = 1$; otherwise, $\varphi_i = 0$. When the i th vector is not a padding node, $\theta_i = 1$; otherwise, $\theta_i = 0$. The compression process of the feature matrix is summarized in Algorithm 1. $D_g \in \mathbb{R}^{n \times l \times d}$ represents the compressed feature matrix, while $M_g \in \mathbb{R}^{n \times l}$ represents the masks for padding vectors in the feature matrix.

Algorithm 1 Compression algorithm of feature matrix.

Input: $D \in \mathbb{R}^{n \times l' \times d}$, $\Phi \in \mathbb{R}^{n \times l'}$, $\Theta \in \mathbb{R}^{n \times l'}$
Output: $D_g \in \mathbb{R}^{n \times l \times d}$, $M_g \in \mathbb{R}^{n \times l}$
 $\eta \leftarrow$ Find the vector for the padding node by Θ .
 $l \leftarrow \max(\text{sum}(\Phi_1), \dots, \text{sum}(\Phi_n))$
 $D_g \leftarrow$ Initialize a tensor of shape (n, l, d) using $\eta \in \mathbb{R}^d$.
 $M_g \leftarrow \mathbf{0}_{n \times l}$
for $i = 1$ to n **do**
 $D'_i \leftarrow$ Extract the feature vectors of heavy atoms from D_i based on Φ_i .
 $D_g[i] \leftarrow D'_i \in \mathbb{R}^{l \times d}$.
Set the value of the first $\text{sum}(\Phi_i)$ of $M_g[i]$ to 1.
end for

3.2.3. Constructing the feature matrix for spatial structure

We utilize RDKit to generate the 3D conformations of the drugs and optimize the 3D conformations using the Merck Molecular Force Field (MMFF) (Halgren, 1996). Then, the x -axis, y -axis and z -axis coordinates of the heavy atoms were extracted from the 3D conformations of the drugs.

The sine and cosine encoding (Vaswani et al., 2017) are used to encode the x -axis, y -axis and z -axis coordinates. We represent the encoding vectors of the x -axis coordinates of the i th drug as $\psi_i(x) = \{\bar{x}_1^i, \bar{x}_2^i, \dots, \bar{x}_v^i\} \in \mathbb{R}^{l \times d}$, where $\bar{x}_v^i \in \mathbb{R}^d$ is the encoding vector of the x -axis coordinate of the i th atom and can be computed as follows:

$$\bar{x}_v^{(j)} = \begin{cases} \sin(\mu \cdot p_i^x \cdot w_v), & \text{if } j = 2v \\ \cos(\mu \cdot p_i^x \cdot w_v), & \text{if } j = 2v + 1 \end{cases} \quad (4)$$

$$w_v = \frac{1}{10000^{2v/d}} \quad (5)$$

where $v = 0, 1, 2, \dots, \frac{d}{2} - 1$, p_i^x represent the x -axis coordinates of the i th atom and μ is the scaling factor for the coordinates.

Then, the encoding vector $\psi(x) \in \mathbb{R}^{n \times l \times d}$ of the x -axis coordinates is added to the compressed feature vector D_g to obtain the feature encoding D_x on the x -axis.

$$D_x = \psi(x) + D_g \quad (6)$$

where $D_x \in \mathbb{R}^{n \times l \times d}$. Similarly, we can also obtain the feature encodings $D_y \in \mathbb{R}^{n \times l \times d}$ and $D_z \in \mathbb{R}^{n \times l \times d}$ on the y-axis and z-axis, respectively. Finally, the feature encodings D_x , D_y and D_z are concatenated together with D_g to form the spatial feature encoding matrix of the drug.

$$\tilde{D}_g = D_g \odot D_x \odot D_y \odot D_z \quad (7)$$

where $\tilde{D}_g \in \mathbb{R}^{n \times 4 \times l \times d}$ and $\odot$ represents the operation of dimensionality expansion and concatenation. The purpose of concatenating D_g is to prevent information loss. Because the spatial structure of drugs obtained through RDKit has limited precision, when the precision is low, adding spatial position encoding will actually result in the loss of original features. So, we decided to retain the drug features that do not have spatial positional encoding added.

3.2.4. Extracting the spatial structural features of drugs

Then, we use spatial attention (Woo et al., 2018) to extract the spatial structure information of the drug.

The purpose of the space attention module is to find which part is important in the space, that is, which areas of the drug may interact with the target. First, we apply max-pooling and average-pooling along the channel axis to extract features with significant changes and global features in the drug space. Then, the two obtained feature maps are connected in the channel dimension. Next, a convolution layer is used to generate a spatial attention map. The spatial attention can be computed as follows:

$$att_s = \sigma(\text{Conv}([\text{MaxPool}(\tilde{D}_g); \text{AvgPool}(\tilde{D}_g)])) \quad (8)$$

where Conv is a convolution operation with the filter size of 5×5 and σ is the sigmoid activation function. Then, apply batch normalization to expedite model convergence. Finally, extract the maximum value on each channel to obtain the spatial structural feature D_p of the drug.

$$D_p = \text{Max}(\text{BatchNorm}(att_s \odot \tilde{D}_g)) \quad (9)$$

where $\odot$ denotes element-wise product and $D_p \in \mathbb{R}^{n \times l \times d}$. Then, we utilized a dimension level gate mechanism to fuse the compressed drug feature matrix with the extracted spatial structural feature.

$$D_s = \alpha \cdot D_g + (1 - \alpha) \cdot D_p \quad (10)$$

where $D_s \in \mathbb{R}^{n \times l \times d}$ represents the spatial feature matrix of the drugs and where α is a trainable parameter.

3.3. Target search network

As shown in Fig. 3, the target search network is composed of a transformer encoder and two decoders. $\mathcal{L}_{cons}$ is the consistency loss function. The input of the encoder is n-gram (Dong et al., 2006; Tsubaki et al., 2019) substrings of amino acid sequence. Therefore, there are about 20^n different residues that can be encoded.

The amino acid sequence determines the spatial structure and function of a target. The encoder module is to explore interactions among the different amino acids. The two decoder modules aim to explore the binding sites between drugs and targets from two dimensions of the drugs: the 2D substructures and the 3D spatial structure. The 'K', 'V' and 'mask' generated by the encoder are used as inputs of the two decoders. The 'mask' is used to distinguish the positions of padding nodes in the targets.

We represent the 'Q' of the i th head of the drug 2D substructures as $Q_d^{(i)} \in \mathbb{R}^{p \times \frac{d}{k}}$, where k is the number of heads of attention and p is the number of drug 2D substructures. $K_t^{(i)} \in \mathbb{R}^{q \times \frac{d}{k}}$ and $M_t \in \mathbb{R}^q$ represent the 'K' and 'mask' of the target, respectively, where q is the length of the target. The search attention scores $A_b^{(i)} \in \mathbb{R}^{p \times q}$ of the i th head between drug 2D substructures and amino acids can be computed as follows:

$$A_b^{(i)} = \text{softmax}(\text{R}(\frac{Q_d^{(i)} K_t^{(i)T}}{\sqrt{d/k}}, M_t)) \quad (11)$$

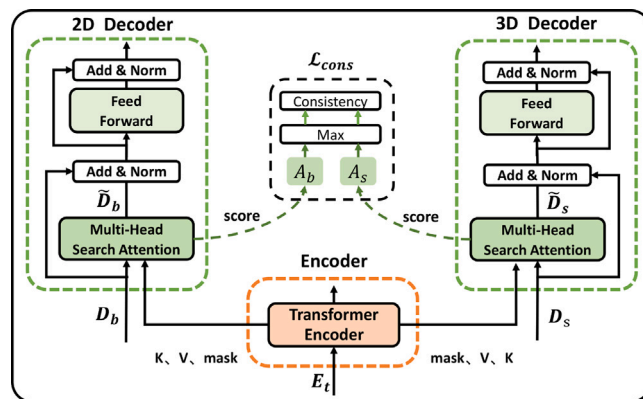


Fig. 3. The schematic of the target search network. $\mathcal{L}_{cons}$ is the consistency loss function.

Similarly, the search attention scores $A_s^{(i)} \in \mathbb{R}^{l \times q}$ of the i th head between drug spatial structure and amino acids can be calculated. We represent the 'V' of the i th head of the target as $V_t^{(i)} \in \mathbb{R}^{q \times \frac{d}{k}}$. The updated features D_b' of the substructures can be computed as follows:

$$D_b' = \text{Concat}(A_b^{(1)} V_t^{(1)}, \dots, A_b^{(k)} V_t^{(k)}) \quad (12)$$

where $D_b' \in \mathbb{R}^{p \times d}$ represents the feature matrix of the 2D substructures of a drug, the feature matrix of the substructures for the entire batch of drugs can be denoted as $\tilde{D}_b \in \mathbb{R}^{n \times p \times d}$. Similarly, the feature matrix $\tilde{D}_s \in \mathbb{R}^{n \times l \times d}$ of the drug spatial structure can be calculated.

After obtaining the outputs of the encoder and decoders, we extract the maximum value on each dimension from the feature vectors. Finally, we use $f_b \in \mathbb{R}^{n \times d}$ and $f_s \in \mathbb{R}^{n \times d}$ to represent the feature vectors of the 2D substructures and spatial structure of the drug, respectively. The feature vector of the target is represented as $f_t \in \mathbb{R}^{n \times d}$.

3.4. Interaction prediction network

The interaction prediction network is a three-layer perceptron. First, we combine the feature vectors f_b of the 2D substructures and f_s of the spatial structure of the drug to obtain the final drug feature vector $f_d \in \mathbb{R}^{n \times d}$. Next, we concatenate the obtained target feature vector f_t with the drug feature vector f_d and feed them into the interaction prediction network to obtain the classification result $O_i \in \mathbb{R}^{n \times 2}$ for DTI and the prediction result $O_a \in \mathbb{R}^n$ for CPA. The formula is as follows:

$$O_i = \text{Linear}(64 * 2, 512, 32, 2) \quad (13)$$

$$O_a = \text{Linear}(64 * 2, 1024, 64, 1) \quad (14)$$

where 64 is the value of the embedding dimension d for drug and target features. The hidden layers utilize the ReLU activation function and the Dropout function.

3.5. Loss function

Since $A_b^{(i)}$ and $A_s^{(i)}$ represent the attention scores between the residues of the target and the substructures and spatial structure of the same drug, their distributions should be consistent. Using $A_b^{(i)}$ as an example, we first extract the maximum attention score for each residue. Next, we aggregate the attention distributions from all heads to derive the final attention distribution $T_j^b \in \mathbb{R}^q$ for the j th drug's 2D substructures on the residues, that is,

$$T_j^b = \text{Max}(A_b^{(1)}) + \text{Max}(A_b^{(2)}) + \dots + \text{Max}(A_b^{(k)}) \quad (15)$$

Similarly, the attention distribution $T_j^s \in \mathbb{R}^q$ of the spatial structure of the j th drug on the residues can be calculated. The purpose of

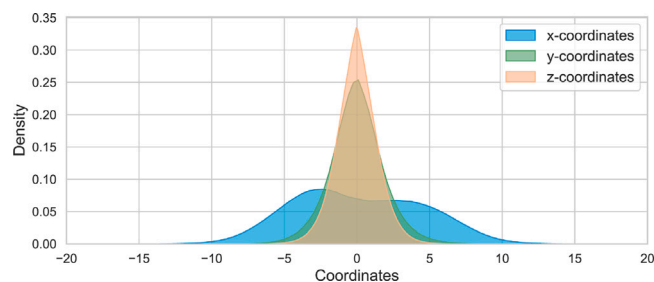


Fig. 4. Distribution plot of spatial coordinates for all small molecules. The x-axis represents the coordinates, and the y-axis represents the density values.

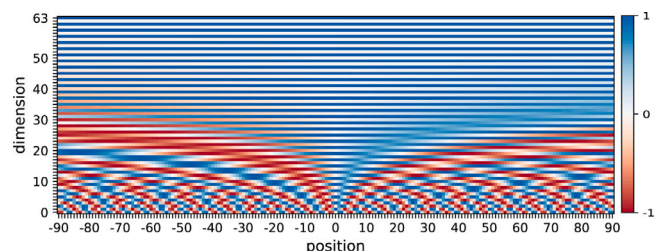


Fig. 5. Visualization of position encoding. The x-axis represents the position coordinates and the y-axis represents the embedded dimensions of positions. The closer the distance between two position coordinates, the more similar their embedded vectors will be.

constructing the consistency loss function is to maximize the similarity between T_j^b and T_j^s as much as possible. Therefore, when the similarity is low, we aim to impose a significant penalty on the model, and when the similarity is high, we apply a smaller penalty to the model. The consistency loss $\mathcal{L}_{cons}$ for the entire batch can be computed as follows:

$$\mathcal{L}_{cons} = \frac{1}{n} \sum_{i=1}^n \left(1 - \frac{T_i^b \cdot T_i^s}{\|T_i^b\| \|T_i^s\|} \right) \quad (16)$$

where n is the batch size and $\mathcal{L}_{cons} \in [0, 2]$.

Since our experiments involve both DTI (binary classification task) and CPA (regression task), we set different loss functions for each task. For DTI, we build the $\mathcal{L}_{DTI}$ by combining the $\mathcal{L}_{cons}$ with the cross-entropy loss.

$$\mathcal{L}_{DTI} = \mathcal{L}_{cons} - \frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (17)$$

where y_i represents the actual interaction case between drug and target and $\hat{y}_i$ represents the predicted probability by the model. For CPA, we build the $\mathcal{L}_{CPA}$ by combining the $\mathcal{L}_{cons}$ with the mean square error, that is,

$$\mathcal{L}_{CPA} = \mathcal{L}_{cons} + \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (18)$$

4. Experiments and discussions

4.1. Implementation and parameter settings

The MddTI model is implemented using PyTorch on an Nvidia Quadro P4000 graphics card with 8 GB of graphics memory. In MddTI, we set the batch size to 16. The embedding dimensions for drug 2D substructures and atoms are both 64. The embedding dimension for the target is 64, and the maximum length for the target is set to 1000. This length is a common choice, as it covers over 90% of the targets in the datasets. Any amino acids exceeding this length will be directly discarded. Since drugs typically have fewer atoms, we do not impose a maximum length limit on them. Increasing the number of attention

heads in the target search network leads to a decrease in the dimension of each head. To ensure an appropriate dimension, we set 2 attention heads with a dimension of 32 each.

When constructing the feature matrix for the spatial structure of drugs, we choose to scale the coordinates proportionally using a scaling factor μ due to the small coordinate values. The distribution of spatial coordinates for all small molecules is depicted in Fig. 4. We observe that the majority of coordinates are located between $-5 \text{ \AA}$ and $5 \text{ \AA}$. According to Fig. 5, it can be observed that this encoding scheme can better distinguish different positions when the coordinate values are between -50 and 50 . Therefore, we set the value of μ to 10.

4.2. Analysis of the number of substructures

We compared and analyzed the number of substructures obtained after decomposing drugs using the ICMF and ESPF algorithms. The statistical results are shown in Fig. 6, revealing that our proposed decomposition method generally yields slightly fewer substructures compared to the ESPF method. The reason for this is that the ESPF algorithm decomposes SMILES, which may generate descriptor fragments like ‘(’, ‘)’ and ‘1’. Since our method is founded on the two-dimensional molecular graph of drugs, it avoids the generation of descriptor fragments.

4.3. Performance on four DTI datasets

We compare our models with some deep learning methods including CPI (Tsubaki et al., 2019), GraphDTA (Nguyen et al., 2021), TransformerCPI (Chen et al., 2020), TrimNet-CNN (Li et al., 2021), DeepEmbedding-DTI (Chen et al., 2021), MINN-DTI (Li et al., 2022b), MolTrans (Huang et al., 2021) and HyperAttentionDTI (Zhao et al., 2022). In the experiments, we use the 5-fold cross-validation method to divide the dataset into training set and test set, and then divide 10% from the training set as the validation set. The final performance evaluation standard of the model is the average value of five experiments.

Table 4 shows the comparison results of the proposed models with baselines on the HUMAN and C.ELEGANS datasets. The drug substructures in MddTI-ICMF are generated using the ICMF algorithm, while the substructures in MddTI-ESPF are generated using the ESPF algorithm. The results indicate that, our models has the highest the Area Under Curve (AUC), the Area Under PR (AUPR) and Recall. In addition, the performance of model using ESPF is better than that using ICMF. Based on the prediction results of MddTI-ICMF, we plotted the ROC and PR curves on the Human and C.elegans datasets. The results are shown in Fig. 7.

The experimental results on the Davis and KIBA datasets are shown in Table 5. The results indicate that, our model achieves the highest score in 3 out of 5 metrics on the KIBA dataset, and outperforms the comparison models in 4 out of 5 metrics on the Davis dataset. However, on the Davis dataset, our models slightly underperform MDTT in terms of the AUC metric. This is due to the limited number of drug categories, with only 68 in this dataset, which potentially has a suppressing effect on our models' performance. In addition, we found that the performance of MddTI-ICMF and MddTI-ESPF is very similar on the both datasets.

4.4. Performance on three CPA datasets

We predicted the binding affinity between compounds and proteins on the Kd, EC50 and Ki datasets. In general, smaller Kd, EC50 and Ki values indicate a stronger binding affinity between small molecules and proteins. The baselines used here include DeepAffinity (Karimi et al., 2019), MONN (Li et al., 2020), DeepPurpose (Huang et al., 2020) and BACPI (Li et al., 2022a).

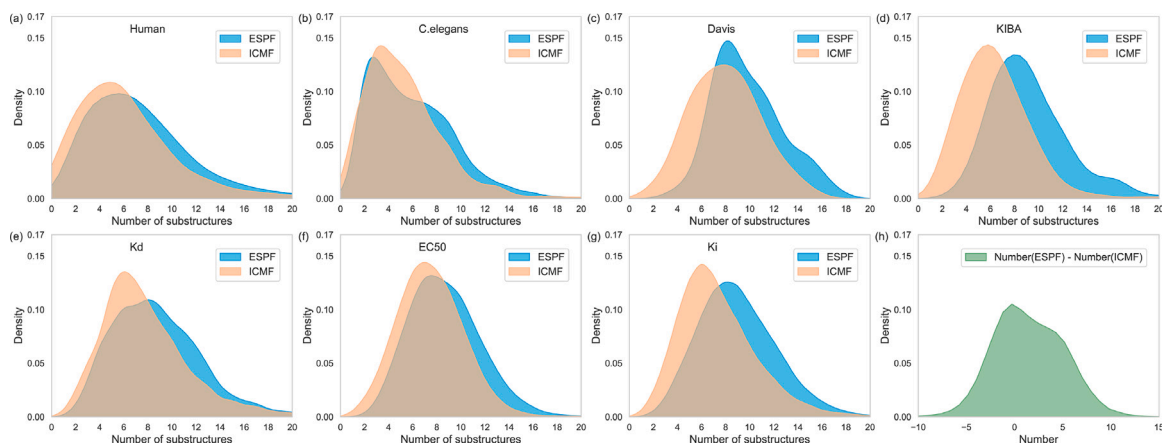


Fig. 6. Probability density distribution plot of the number of substructures. (a)–(g) represent the probability density distribution plots of the number of substructures obtained through the ESPF and ICMF methods on the seven datasets, respectively. (h) represents the probability distribution plot of the difference between the number of substructures obtained by ESPF and ICMF methods.

Table 4

Performance comparison between our model and existing methods on the random splitting Human and C.elegans datasets.

Model	Human			C.elegans		
	AUC	Precision	Recall	AUC	Precision	Recall
CPI ^a	0.970	0.918	0.923	0.978	0.938	0.929
GraphDTA ^a	0.960 ± 0.005	0.882 ± 0.040	0.912 ± 0.040	0.974 ± 0.004	0.927 ± 0.015	0.912 ± 0.023
TransformerCPI	0.973 ± 0.002	0.916 ± 0.006	0.925 ± 0.006	0.988 ± 0.002	0.952 ± 0.006	0.953 ± 0.005
TrimNet-CNN	0.974	0.918	0.953	0.987	0.946	0.945
MINN-DTI	0.981 ± 0.003	0.945 ± 0.030	0.902 ± 0.045	–	–	–
MdDTI-ICMF (ours)	0.987 ± 0.001	0.939 ± 0.011	0.951 ± 0.005	0.991 ± 0.002	0.962 ± 0.007	0.960 ± 0.004
MdDTI-ESPF (ours)	0.989 ± 0.001	0.956 ± 0.004	0.947 ± 0.005	0.994 ± 0.001	0.964 ± 0.004	0.970 ± 0.004

^a Results obtained from Chen et al. (2020).

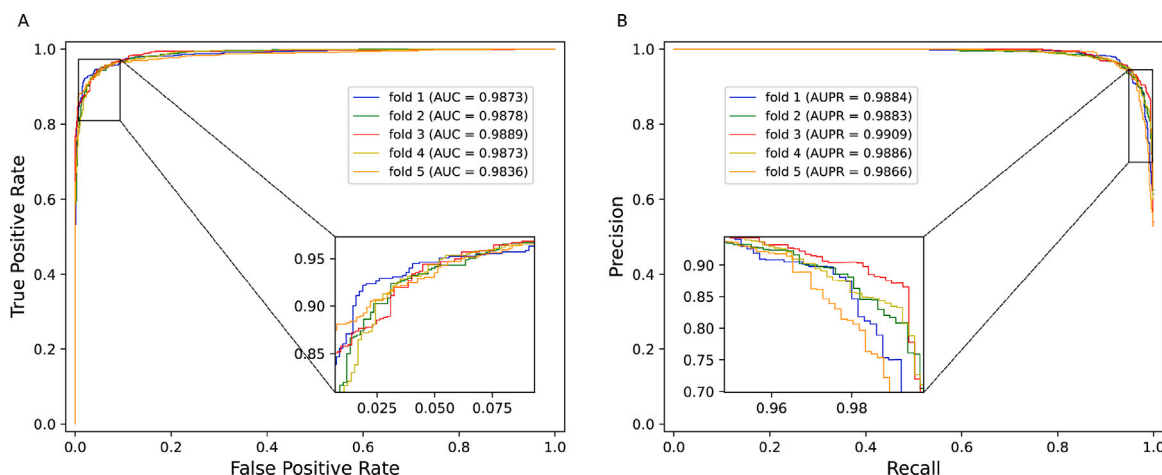


Fig. 7. The ROC and PR curves of MdDTI-ICMF on the Human dataset. The A is the ROC curves and B is the PR curves.

Table 6 shows the Pearson correlation coefficient (PCC) and root mean square error (RMSE) of the baselines on the three datasets. The experimental results indicate that our models perform the best on the Kd and EC50 datasets, with MdDTI-ICMF outperforming MdDTI-ESPF in the three affinity datasets. The drawback is that MdDTI-ICMF performs slightly lower than BACPI on the Ki dataset. Fig. 8 illustrates the real affinity against the predicted value on the three affinity datasets. The predicted results shown in the figure are obtained using MdDTI-ICMF. The closer each point in the graph is to the dashed line, the better the predicted results. Note that the affinity values in the CPA datasets are transformed into log space. For example, the K_i value in

the Ki dataset is replaced with a logarithmic function (pK_i), which is calculated as follows:

$$pK_i = -\log_{10} \left(\frac{K_i}{10^9} \right) \quad (19)$$

4.5. Ablation study

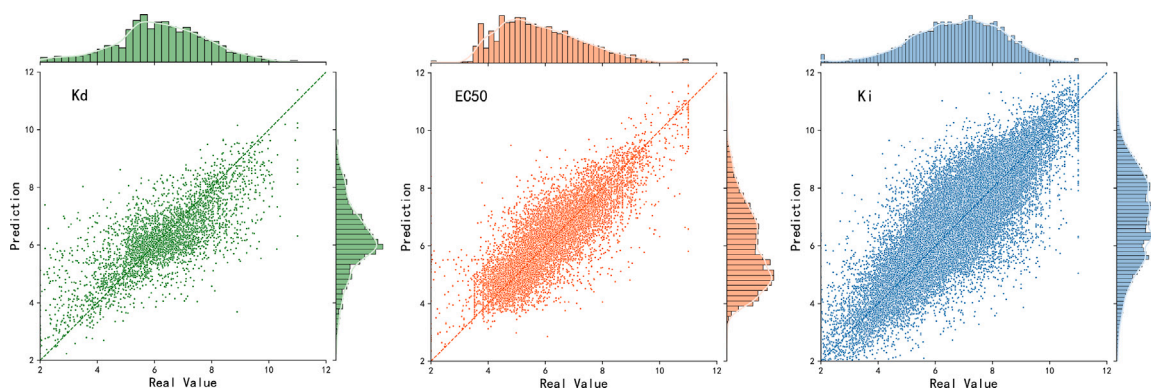
To verify the effect of different modules on the performance of the model, we also conducted four groups of control experiments:

1. –ICMF: removing the 2D substructures of the drugs.

Table 5

The performance of our models on the Davis and KIBA datasets.

Datasets	Model	ACC	AUC	AUPR	Precision	Recall
Davis	CPI ^a	0.819 ± 0.001	0.863 ± 0.001	0.745 ± 0.002	0.731 ± 0.002	0.570 ± 0.002
	GraphDTA ^a	0.817 ± 0.001	0.859 ± 0.004	0.743 ± 0.007	0.743 ± 0.014	0.530 ± 0.017
	TransformerCPI ^a	0.822 ± 0.001	0.877 ± 0.001	0.767 ± 0.001	0.688 ± 0.003	0.688 ± 0.003
	DeepEmbedding-DTI ^a	0.836 ± 0.008	0.878 ± 0.011	0.775 ± 0.020	0.760 ± 0.017	0.618 ± 0.024
	MolTrans ^a	0.842 ± 0.000	0.900 ± 0.001	0.784 ± 0.002	0.782 ± 0.003	0.617 ± 0.004
	HyperAttentionDTI	0.866 ± 0.001	0.920 ± 0.001	0.839 ± 0.001	0.754 ± 0.002	0.780 ± 0.001
	MdDTI-ICMF (ours)	0.869 ± 0.003	0.918 ± 0.001	0.845 ± 0.002	0.795 ± 0.005	0.729 ± 0.009
	MdDTI-ESPF (ours)	0.866 ± 0.002	0.916 ± 0.001	0.845 ± 0.001	0.787 ± 0.005	0.731 ± 0.030
KIBA	CPI ^a	0.867 ± 0.002	0.864 ± 0.005	0.673 ± 0.005	0.727 ± 0.002	0.477 ± 0.007
	GraphDTA ^a	0.889 ± 0.003	0.914 ± 0.001	0.776 ± 0.007	0.775 ± 0.020	0.594 ± 0.032
	TransformerCPI ^a	0.870 ± 0.001	0.888 ± 0.001	0.708 ± 0.001	0.669 ± 0.003	0.631 ± 0.003
	DeepEmbedding-DTI ^a	0.878 ± 0.002	0.889 ± 0.003	0.727 ± 0.006	0.741 ± 0.005	0.556 ± 0.016
	MolTrans ^a	0.881 ± 0.001	0.905 ± 0.001	0.708 ± 0.003	0.710 ± 0.003	0.645 ± 0.003
	HyperAttentionDTI	0.893 ± 0.001	0.935 ± 0.001	0.814 ± 0.002	0.689 ± 0.001	0.796 ± 0.002
	MdDTI-ICMF (ours)	0.907 ± 0.001	0.938 ± 0.001	0.832 ± 0.001	0.803 ± 0.005	0.697 ± 0.040
	MdDTI-ESPF (ours)	0.909 ± 0.001	0.939 ± 0.001	0.836 ± 0.001	0.804 ± 0.004	0.710 ± 0.005

^a Results obtained from Zhao et al. (2022).**Fig. 8.** The real affinity against the predicted value on the CPA datasets. For each point, the x -axis and y -axis represent its true value and predicted value, respectively. The distance from each sample point to $y = x$ represents the discrepancy between its predicted affinity and the real value.**Table 6**

The PCC and RMSE of the baselines on the Kd, EC50 and Ki datasets.

Model	Kd		EC50		Ki	
	PCC ↑	RMSE ↓	PCC ↑	RMSE ↓	PCC ↑	RMSE ↓
DeepAffinity ^a	0.68	1.17	0.83	0.85	0.79	0.96
MONN ^a	0.74	1.07	0.85	0.79	0.83	0.88
DeepPurpose ^a	0.69	1.16	0.85	0.80	0.81	0.92
BACPI	0.73	1.08	0.85	0.78	0.86	0.80
MdDTI-ICMF (ours)	0.751	1.058	0.862	0.778	0.854	0.862
MdDTI-ESPF (ours)	0.743	1.078	0.851	0.806	0.848	0.869

^a Results obtained from Li et al. (2022a). There are four datasets in the original paper, and we chose the three smallest datasets for the experiments.**Table 7**

Ablation studies on the Kd and EC50 datasets.

Model	Kd		EC50	
	PCC ↑	RMSE ↓	PCC ↑	RMSE ↓
MdDTI-ICMF	0.751	1.058	0.862	0.778
-ICMF ^a	0.737	1.072	0.860	0.793
-3D ^a	0.719	1.116	0.840	0.838
- $\mathcal{L}_{cons}$	0.736	1.085	0.863	0.773
-CA	0.749	1.069	0.858	0.786

^a Once the 2D substructures or spatial structure are removed, $\mathcal{L}_{cons}$ ceases to exist.

2. -3D: removing the spatial structures of the drugs.
3. - $\mathcal{L}_{cons}$: removing the consistency loss function.
4. -CA: removing the compression algorithm (removing all hydrogen atoms from the drugs during preprocessing).

The experimental results on the Kd and EC50 datasets are shown in Table 7. The results indicate that incorporating the spatial structural features of drugs significantly improves the performance of the model, while removing the 2D substructures of drugs reduces the performance of the model. Removing the consistency loss function can improve the performance of the model on individual datasets, but the robustness of the model is compromised. The use of the $\mathcal{L}_{cons}$ enables the model to maintain high performance while also exhibiting good generalization capability. In addition, our experimental results indicate that directly removing hydrogen atoms from the drugs reduces the performance of the model. Therefore, a good approach is to first allow information propagation between all atoms in the drug using GNN, and then remove the features of hydrogen atoms using compression algorithms. This approach maintains the model's performance while speeding up the training process.

Furthermore, we also investigated the impact of the scaling factor μ under different values on the model performance. The experimental results of MdDTI-ICMF on the Davis dataset are shown in Table 8. The experimental results indicate that the model performs best when the scaling factors μ are set to 10 and 14, respectively. If the scaling factor is too small, it will not be possible to fully explore the spatial structural features of the drug. An excessively large scaling factor may lead to distortion in the spatial structure of the drug.

4.6. Case study

To explore how our model can effectively learn the interactions between drugs and targets, we selected 2-Nonyl-4-quinolinol 1-oxide (QNO, Drugbank ID: DB08453) and human Cytochrome b-c1 complex

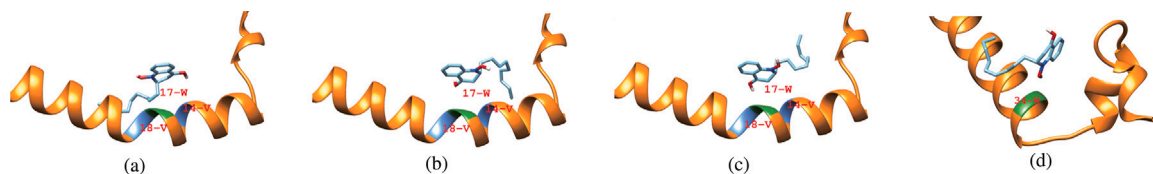


Fig. 9. Docking results of 2-Nonyl-4-quinolinol 1-oxide and Cytochrome b-c1 complex subunit 10. The free energy values for the docking results in figures (a)–(d) are -5.1 kcal/mol, -4.9 kcal/mol, -4.8 kcal/mol and -4.7 kcal/mol, respectively. Lower free energy indicates stronger binding affinity.

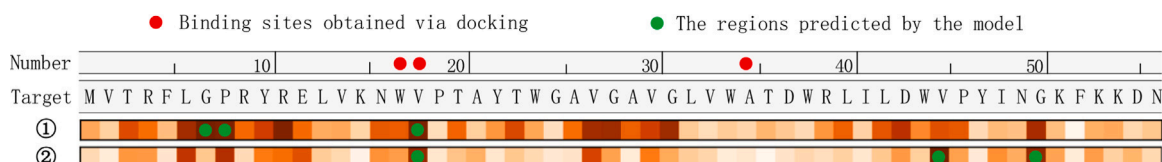


Fig. 10. The attention distribution of QNO on the amino acid sequence of human Cytochrome b-c1 complex subunit 10. ① and ② represent the attention distributions T_{qno}^s of the spatial structure of QNO on the target sequence, respectively. The red dots represent binding sites obtained from docking results, while the green dots represent regions where the models predict higher attention of QNO towards the target.

Table 8

Experimental results of MdDTI-ICMF with different scaling factors μ on the Davis dataset.

	ACC	AUC	AUPR	Precision	Recall
$\mu = 2$	0.867	0.916	0.846	0.790	0.711
$\mu = 6$	0.864	0.915	0.839	0.795	0.709
$\mu = 10$	0.869	0.918	0.845	0.795	0.729
$\mu = 14$	0.865	0.918	0.846	0.784	0.730
$\mu = 18$	0.862	0.917	0.842	0.778	0.725

subunit 10 (Uniprot ID: O14957) for analysis. 2-Nonyl-4-quinolinol 1-oxide is a mitochondrial respiratory chain inhibitor that can affect cellular energy metabolism processes. Cytochrome b-c1 complex subunit 10 plays a crucial role in the mitochondrial respiratory chain, helping to maintain cellular energy metabolism.

We downloaded the spatial structure file of the human Cytochrome b-c1 complex subunit 10 predicted by AlphaFold (Jumper et al., 2021). Subsequently, we performed molecular docking between this structure and QNO using AutoDock Vina (Trott and Olson, 2010). AutoDock Vina is a commonly used docking tool that can predict the potential binding modes between the small molecule and the target. Generally, when the docking binding energy is below -5 kcal/mol, it can be considered that the docking result is relatively stable.

The Fig. 9 displays four optimal docking results with docking scores of -5.1 kcal/mol, -4.9 kcal/mol, -4.8 kcal/mol and -4.7 kcal/mol, respectively. The results indicate a potential interaction between QNO and the 17th and 18th amino acids of the target. Additionally, the 34th amino acid of the target also exhibits some attraction towards QNO. Then, we extracted the attention distribution T_{qno}^s of the spatial structure of QNO towards the target and visualized it. As shown in Fig. 10, heatmaps ① and ② represent the attention distributions T_{qno}^s extracted from MdDTI-ICMF and MdDTI-ESPF, respectively. The green dots in the figure represent 3 positions with higher attention towards the target. The red dots in the figure represent the binding sites obtained based on the docking results. The results indicate that our models successfully predicted the binding site at the 18th residue of the target. Note that we did not extract the attention distribution T_{qno}^b of the 2D substructure of QNO towards the target because the distributions of T_{qno}^b and T_{qno}^s are highly consistent under the regulation of the consistency loss function $\mathcal{L}_{cons}$.

In addition, to validate the potential of the model in drug repurposing, we predicted the binding affinity between the SARS-CoV2 3C-like

Protease (3CL^{Pro}) target and 86 existing drugs. The selected drugs include 82 antiviral drugs (Huang et al., 2020), 1 antiparasitic drug (Ivermectin) and 3 unrelated drugs (Cefazolin, Amoxicillin, Penicillin). One commonly used indicator for predicting drug–target binding affinity is the inhibition constant (Ki), so we chose to use the models trained on the Ki dataset for our predictions.

We used the MdDTI-ICMF and MdDTI-ESPF models trained on the Ki dataset for predictions, and the results are summarized in Table 9. The experimental results indicate that out of the top 5 drugs predicted by the MdDTI-ICMF model, 2 have been clinically proven to exhibit inhibitory effects on 3CL^{Pro}. Furthermore, out of the top 5 drugs predicted by the MdDTI-ESPF model, 4 have been successfully predicted. The Ki values between 3 unrelated drugs and 3CL^{Pro} predicted by two models are all very large, indicating that the strength of their interactions is very weak. This indicates that our models can play a role in drug repurposing. Additionally, we found that the predictions from MdDTI-ICMF are more daring, while the predictions from MdDTI-ESPF are more concentrated.

5. Conclusion

In this work, we introduce MdDTI, a novel drug–target interaction prediction model. Unlike previous methods, we simultaneously utilized the 2D substructure and spatial structure of the drug. Compared to the algorithm that decompose SMILES based on the frequency of substring occurrences, the decomposition strategy based on the 2D molecular graph of the drug can reduce some descriptor fragments. When constructing the spatial expression matrix of the drug, we only select to extract the spatial Cartesian coordinates of heavy atoms. For hydrogen atoms, we choose to first use GNNs to transfer their information to the neighboring heavy atoms, and then discard the hydrogen atoms. This approach can help accelerate the training process while maintaining the model's performance. Extensive experimental results indicate that our model is capable of capturing the potential binding regions between drugs and targets, and it shows efficacy in drug repurposing.

Although MdDTI is capable of effectively extracting the spatial structural features of drugs, it still has limitations in handling proteins. Currently, there are already models available for predicting protein structures, such as AlphaFold. Additionally, there are language models for predicting protein functions, such as ESM-1b (Rives et al., 2021). In the future, we will explore how to incorporate the three-dimensional structure information of targets into our current methods in a meaningful way.

Table 9

Predicted binding affinities between existing drugs and SARS-CoV2 3C-like Protease using MdDTI-ICMF and MdDTI-ESPF.

MdDTI-ICMF				MdDTI-ESPF			
Rank	Drug	Affinity	Source	Rank	Drug	Affinity	Source
1	Grazoprevir	7.19		1	Glecaprevir	14.92	Shamsi et al. (2020)
2	Ganciclovir	21.7		2	Etravirine	16.04	
3	Glecaprevir	30.36	Shamsi et al. (2020)	3	Darunavir	18.10	Costanzo et al. (2020)
4	Darunavir	85.15	Costanzo et al. (2020)	4	Ivermectin	30.28	López-Medina et al. (2021)
5	Etravirine	105.86		5	Tipranavir	35.94	Dayer (2020)
69	Cefazolin	11 640.80		29	Cefazolin	1931.92	
84	Amoxicillin	168 643.82		61	Amoxicillin	22 562.19	
85	Penicillin	270 827.50		67	Penicillin	39 641.69	

Note : The model predicts the value of pK_i . The affinity strength K_i in the table is calculated by $10^{-pK_i} \times 10^9$. A smaller K_i value indicates a stronger binding affinity.

CRediT authorship contribution statement

Huaihu Li: Conceptualization, Methodology, Software, Investigation, Visualization, Writing – original draft. **Shunfang Wang:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Weihua Zheng:** Writing – review & editing. **Li Yu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work is supported by the National Natural Science Foundation of China (62062067) and the Program of Yunnan Key Laboratory of Intelligent Systems and Computing, China (202205AG070003).

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